Mathematics and Statistics

Usually the biggest block of the paper. Subtopics listed are typical coverage, not an analysis of a specific paper.

Core Topics

Matrices and Determinants

- Types of matrices, addition, multiplication, transpose
- Determinant properties and evaluation
- Inverse, adjoint, and rank of a matrix
- Solving linear equations (Cramer's rule, matrix method)

Calculus

- Limits and continuity
- Derivatives: rules, chain rule, standard results
- Integration: standard formulas, substitution, by parts, definite integrals
- Maxima and minima, and basic applications

Probability and Basic Statistics

- Mean, median, and mode
- Probability basics and conditional probability
- Bayes' theorem
- Distributions (binomial, Poisson, normal)

Sets, Relations, Functions, and Permutations/Combinations

- Set operations and Venn diagrams
- Types of relations (reflexive, symmetric, transitive, equivalence)
- Functions: one-one, onto, inverse, composition
- Permutations and combinations (nPr, nCr) and counting problems

Coordinate Geometry, Sequences and Series, and Vectors

- Straight lines, circles, and basic conics
- Arithmetic and geometric progressions, and their sums
- Vector algebra: dot product and cross product
- Distance, section formula, and slopes